Inflammopharmacology (2024) 32:101–125
https://doi.org/10.1007/s10787-023-01318-6

Inflammopharmacology

REVIEW

Natural agents as wound-healing promoters

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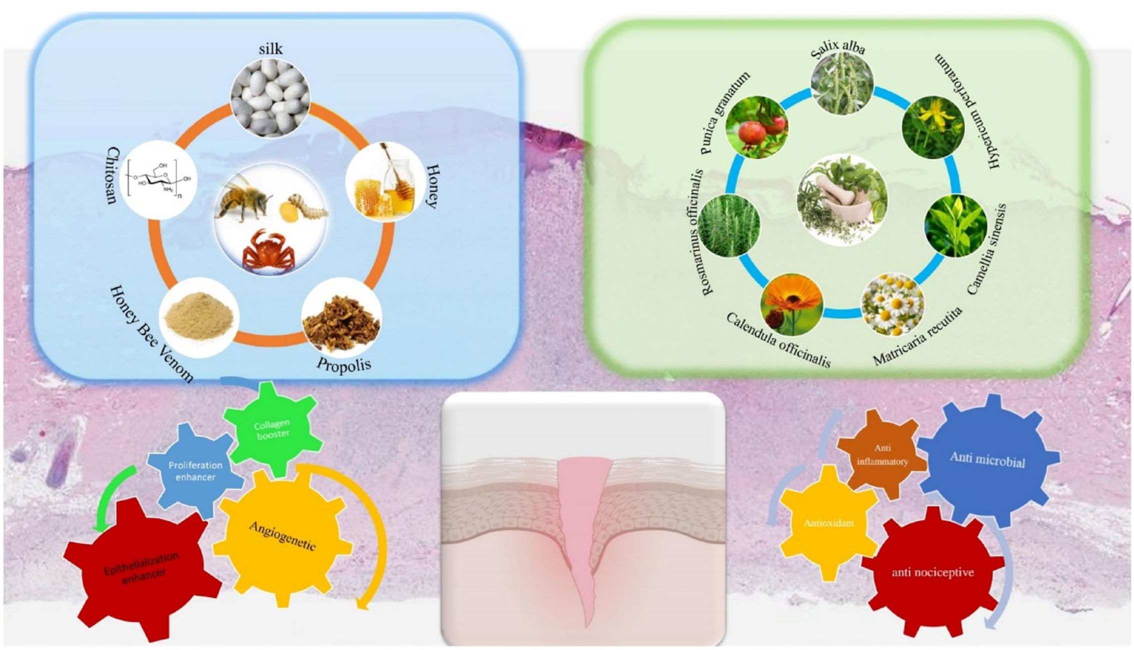
Received: 8 July 2023 / Accepted: 8 August 2023 / Published online: 7 December 2023

Abstract

The management of acute and chronic wounds resulting from diverse injuries poses a significant challenge to clinical practices and healthcare providers. Wound healing is a complex biological process driven by a natural physiological response. This process involves four distinct phases, namely hemostasis, inflammation, proliferation, and remodeling. Despite numerous investigations on wound healing and wound dressing materials, complications still persist, necessitating more efficacious therapies. Wound-healing materials can be categorized into natural and synthetic groups. The current study aims to provide a comprehensive review of highly active natural animal and herbal agents as wound-healing promoters. To this end, we present an overview of in vitro, in vivo, and clinical studies that led to the discovery of potential therapeutic agents for wound healing. We further elucidated the effects of natural materials on various pharmacological pathways of wound healing. The results of previous investigations suggest that natural agents hold great promise as viable and accessible products for the treatment of diverse wound types.

Graphical Abstract



Keywords Wound healing · Wound repair · Natural products · Herbal agents · Animal agents

Extended author information available on the last page of the article

Abbreviations

BV	Bee venom
IL	Interleukin
MDA	Malondialdehyde
MMP	Matrix metalloproteinase
MRSA	Methicillin-resistant Staphylococcus aureus
NF-κB	Nuclear factor-κB
Nrf2	Nuclear factor erythroid 2-related factor 2
PDGE2	Palladogermanide
PDGF	Platelet-derived growth factor
PGE2	Prostaglandin E2
ROS	Reactive oxygen species
SCS	Sulfonated chitosan
SOD	Superoxide dismutase
TGF β	Transforming growth factor β
TNF	Tumor necrosis factor
VEGF	Vascular endothelial growth factor

1. Introduction

As the integumentary system, the skin is the largest and most exposed organ of the human body, performing a wide range of functions. It comprises multiple layers that serve as a protective barrier, shielding the body against various environmental insults, microorganisms, light, and toxic substances. Additionally, the skin plays a vital role in vitamin D synthesis, contributing to thermoregulation and immune surveillance. However, the skin's structural and functional integrity can be compromised by various internal or external damaging parameters, such as cuts, burns, surgical incisions, as well as acute and chronic wounds resulting from physical and mechanical injuries (Han and Ceilley 2017; Ibrahim et al. 2018; Ojeh et al. 2015; Sanford and Gallo 2013; Liu et al. 2022).

Retrospective statistical analyses have demonstrated that millions of individuals suffer from various wound types annually, resulting in substantial healthcare costs associated with their management. Therefore, it is imperative to understand the skin's physiology, wound classification, different stages of the repair process, and identifying factors that may impede the natural course of wound healing (Sen 2019).

The wound-healing process is a complex and dynamic biological process involving multiple cell types' coordinated efforts and molecular signaling pathways. The process can be divided into four distinct but overlapping phases: hemostasis, inflammation, proliferation, and remodeling. The inflammatory phase is characterized by vascular responses that result in coagulation and hemostasis of blood, as well as cellular infiltration, primarily by leukocytes with diverse functions. Hemostasis is a distinct phase that precedes inflammation as it relates to vascular responses to prevent further blood loss. The proliferative phase is marked by the formation of new epithelium, which covers the wound surface, and the simultaneous growth of granulation tissue that fills the wound space and involves fibroblast proliferation, extracellular matrix deposition, and new blood vessel formation. Finally, in the remodeling phase, the newly formed tissue is remodeled and strengthened to restore tissue integrity. Although these phases occur sequentially, they may overlap in time, making the wound-healing process a dynamic and intricate (Li et al. 2007).

Impaired wound healing is common, particularly in chronic and delayed acute wounds, which often fail to progress beyond the inflammatory phase. This delay is attributed to various factors that impair and postpone the healing process. Non-healing wounds pose a significant health and economic burden on society, particularly among individuals aged 65 years and older, who are at an increased risk of developing age-related diseases such as diabetes (Guo and DiPietro 2010; Dreifke et al. 2015). Moreover, non-healing wounds often lead to complications that impede the healing process, including patient infections (Menke et al. 2007).

Various materials have been developed to address these challenges to protect wounds from external agents, accelerate closure by maintaining wound moisture, and minimize scarring. These materials include natural and synthetic fibers, polymers, colloids, gel-forming agents, and hydrogels, which can be applied as wound dressings, composite materials, and skin regeneration scaffolds (Saghazadeh et al. 2018a, b). This review critically evaluates the recent progress and advancements in various areas of wound care using natural agents. To this end, the wound-healing stages are discussed, and factors affecting wound healing are briefly described. Finally, the natural wound-healing materials derived from animal and plant sources will be reviewed in detail.

Wound-healing process

Damaged skin can restore its integrated structure and normal functions by collaborating with different cell types, cytokines, mediators, and the vascular system, which trigger several biological processes. As shown in Fig. 1, the biological phases of wound healing can be expressed in four distinct stages: the hemostatic phase, inflammatory reactions, cell proliferation (granulation), and maturation (remodeling phase).

Homeostasis

During the hemostatic phase of wound healing, which occurs within the first few minutes to hours after injury,

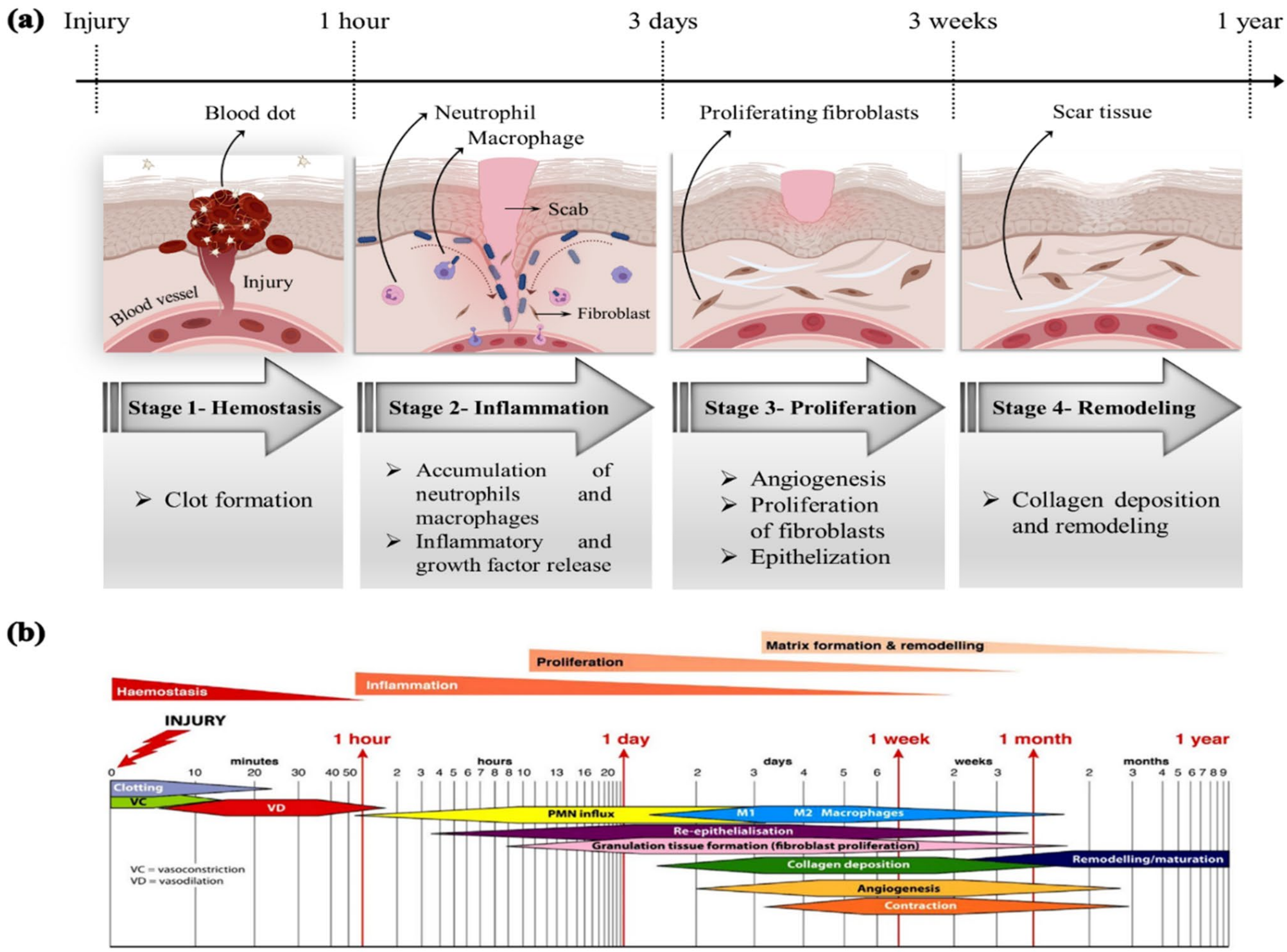


Fig. 1 The four stages of the wound-healing process and the corresponding time scale. (**a**) Cellular influences in wound healing [reprinted with permission from (Elham Asadian 2022)], and (**b**) overlapping four phases of wound healing and their time scales (Greaves et al. 2013)

activated platelets in the blood interact with fibrin to form a clot (Gonzalez et al. 2016; Saghazadeh et al. 2018a, b; Tottoli et al. 2020).

Inflammation

The initiation of the inflammatory phase during tissue repair is marked by the release of signaling molecules from platelets and damaged cells. These mediators trigger the activation and infiltration of inflammatory cells, particularly neutrophils, and macrophages, to the wound lesion. During this stage, the release of lysosomal enzymes, growth factors, and immune cells, as well as the generation of abundant reactive oxygen species (ROS), induces pain, swelling, edema, and erythema in the wound site (Gonzalez et al. 2016; Saghazadeh et al. 2018a, b; Tottoli et al. 2020).

Proliferation

The proliferation stage of wound healing involves the formation of granulation tissue, re-establishing the new blood vessel network, and re-epithelialization, resulting in the generation of new collagen and extracellular matrix. This process is driven by the proliferation and migration of various cells, including fibroblasts, keratinocytes, and endothelial cells. These cells interact with each other, forming complex signaling networks that regulate cell behavior, such as cell proliferation, migration, and differentiation. Additionally, angiogenesis, which is the formation of new blood vessels, is critical in providing nutrients and oxygen to the newly forming tissues. This phase is characterized by the deposition of extracellular matrix components, such as collagen and elastin, which support the growth and remodeling of new tissue (Gonzalez et al. 2016; Saghazadeh et al. 2018a, b; Tottoli et al. 2020).

Remodeling

The final phase of wound healing, known as remodeling, typically commences several weeks after the initial injury and may extend up to 2 years. The primary objective of this phase is to reinstate the normal architecture of the skin, collagen, and extracellular matrix, resulting in scar tissue formation. Failure to regulate any of these pathways can significantly impact the wound-healing process. Multiple factors, such as infection, necrosis, trauma, edema, aging, chronic diseases (e.g., diabetes), and drug or disease-induced immunosuppression, can delay the process of wound healing (Gonzalez et al. 2016; Saghazadeh et al. 2018a, b; Tottoli et al. 2020). In cases where a large area of skin is damaged, a range of treatments and follow-up care may be required to prevent severe bleeding, microbial infections, and intense pain, as well as to expedite the wound-healing process. The time scale of each stage is depicted in Fig. 1b (Greaves et al. 2013).

The utilization of natural compounds possessing medicinal properties has a rich historical background. Natural compounds have gained recognition for their biocompatibility, anti-inflammatory, antimicrobial, and antioxidant activities, making them suitable for various clinical applications, particularly in skin repair. Natural compounds have been shown to influence one or more stages of wound healing, resulting in more effective wound care management for dermatological disorders. This review aims to focus on naturally derived substances and their mechanisms of action, which have several advantages for the wound healing. To achieve this goal, we comprehensively analyze relevant publications and highlight the cellular influences in wound healing. In particular, we examine the potential of natural compounds in promoting wound healing.

Factors affecting wound healing

Numerous factors can impede and prolong the wound-healing process. These factors can be classified into two groups: local and systemic factors. Local factors directly affect wound characteristics, while systemic factors refer to the patient's overall health state that can influence the healing process (Guo and DiPietro 2010).

Local factors

Oxygenation

Adequate oxygen supply is crucial for cellular metabolism, specifically in energy generation via ATP, and plays a significant role in wound healing. Oxygen is essential to impede infection, promote angiogenesis, and enhance migration, re-epithelialization, collagen synthesis, and proliferation, ultimately resulting in a more effective wound-healing process (Rodriguez et al. 2008; Bishop 2008).

Infections

Microbial colonization on the skin surface is common, and microorganisms may penetrate underlying tissues upon skin injury. The replication state of microorganisms determines whether the wound is infected: non-replicating microorganisms on the injury site lead to contamination, while replicating microorganisms on the wound without tissue damage is called colonization. As a local stage, infection colonization and microorganism replication may initiate local tissue responses, thereby serving as an intermediate stage of wound healing (Edwards and Harding 2004).

Systemic factors

Age

As per the World Health Organization (WHO) report, the global population of individuals aged 60 and above is growing faster than any other age group. Studies conducted on animal models and clinical trials have indicated that this age group is at an increased risk of impaired wound healing (Keylock et al. 2008; Gosain and DiPietro 2004).

Sex hormones

Various endogenous hormones can influence the wound-healing process, including male androgens (such as testosterone), female hormones (like estrogens), and the steroid precursor. The effects of these hormones on wound healing have been extensively studied, and it has been shown that they play an important role in the process. For instance, aged males have demonstrated a greater delay in acute wound healing than younger males (Gilliver et al. 2007).

Stress

Research studies have demonstrated that psychological stress may impede the wound-healing process in animals and humans. For instance, individuals who have Alzheimer's and students experiencing stress during examinations have been shown to exhibit delayed wound healing (Marucha et al. 1998).

Diabetes

Patients with diabetes mellitus exhibit a delay in the acute wound-healing process and are at risk of developing chronic non-healing diabetic foot ulcers, which can lead to serious complications (Brem and Tomic-Canic 2007).

Medications

The effects of various medications on wound healing are well documented. Medications that interfere with clot formation, cell proliferation, and inflammatory responses have been shown to impact the wound-healing process significantly. For example, chemotherapeutic drugs can delay cell migration to the wound, reduce collagen synthesis, hinder fibroblast proliferation, and impede wound contraction, ultimately leading to impaired wound healing. It is important to consider medication use when evaluating a patient's wound healing potential and to manage their medication regimen accordingly (Franz et al. 2007).

Obesity

Obese individuals undergoing surgical procedures have demonstrated a higher incidence of surgical-site infections and wound complications than their non-obese counterparts. This can be attributed to several factors, including reduced delivery of antibiotics, hypoperfusion, and ischemia in subcutaneous adipose tissue. Hence, obese patients require specialized wound care management to prevent delayed or impaired wound healing ('The Obese Surgical Patient: A Susceptible Host for Infection' 2006).

Alcohol and smoking

Based on animal and human studies, alcohol consumption interferes with and delays wound healing and increases the risk of infection (Szabo and Mandrekar 2009). In the post-operative period, smokers exhibit a delay in wound healing and an increased incidence of complications, such as infection, wound necrosis, and dehiscence, resulting in decreased wound tensile strength (Ahn et al. 2008).

Nutrition

The role of nutrition is widely recognized as a critical factor in the wound-healing process. Protein deficiency, for instance, can lead to impairment of the remodeling phase, disturb capillary formation and fibroblast proliferation, and interfere with proteoglycan and collagen synthesis. Vitamins C, A, and E also show antioxidant and anti-inflammatory activities that can facilitate wound healing (Campos et al. 2008). In addition, lipids, such as fatty acids, may be utilized as nutritional supplements to fulfill the energy requirements and provide essential building blocks for tissue repair in wound healing (McDaniel et al. 2008).

Wound-healing agents

The materials applied in wound-healing applications must possess specific properties to achieve optimal outcomes. Biocompatibility is considered one of the most important characteristics, as it plays a critical role in eliciting an appropriate host response. Recent studies have suggested using bioactive materials due to their ability to interact with biological environments and positively impact cellular behavior (Catanzano and Boateng 2020).

Wound-healing agents can generally be classified into natural and synthetic categories (Negut et al. 2020). Natural agents are typically derived from natural sources, such as animals and plants, and have demonstrated many advantages in biomedical applications compared to synthetic ones, including biodegradability, biocompatibility, and biological activity. These materials offer a promising platform as a substitute for the natural structural components of the skin background (Portela et al. 2019).

Animal products for wound-healing

Bee products

Bee venom Apitherapy is an ancient therapeutic approach that remains widely utilized today. Bee venom, derived from *Apis mellifera*, is a bee product that can be obtained through the non-lethal stunning electric method (Han et al. 2011). Bee venom (BV) comprises over 18 biologically active compounds that exhibit significant potential in the treatment of different diseases. The major constituents of bee venom include Melittin, which accounts for 50% of its total dry weight, as well as adolapin and apamin (Al-Waili et al. 2015; Baqer and Yaseen 2018b; Kim et al. 2019). Several studies have investigated the mechanisms and functions of BV in wound healing, with a selection of these findings presented herein.

A prolonged inflammatory phase can have detrimental effects on the wound-healing process. BV has been shown to reduce cytokines and inflammatory mediators, such as interleukins (IL-1, IL-6, IL-8), tumor necrosis factor (TNF), nuclear factor-κB (NF-κB), and palladogermanide (PGE2), thereby suppressing inflammation, and reveals antinociceptive properties (Al-Waili et al. 2015). In this regard, melittin and adolapin help inhibit prostaglandins and cyclooxygenase, respectively. Besides, tertiapin supports the anti-inflammatory effects of other BV peptides by blocking potassium channels (Kurek-Górecka et al. 2021). Additionally, BV has been found to play a significant role in elevating collagen levels, accelerating wound healing, and reducing scarring. BV helps prevent collagen degradation by inhibiting matrix metalloproteinase-1 (MMP1) and matrix metalloproteinase-3 (MMP3) and enhances collagen synthesis by increasing the expression of hydroxyproline, the main amino acid involved in collagen production. Studies have demonstrated that type I collagen levels have increased in normal, chronic, and diabetic wounds following bee venom therapy (Han et al. 2011; Al-Waili et al. 2015; Hozzein et al. 2018).

By promoting glutathione, one of the body's main natural antioxidant compounds, BV inhibits free radicals and helps eliminate destructive oxidative compounds. Also, melittin is indicated to inhibit lipid peroxidase and nitric oxide; consequently, BV's antioxidant properties may also contribute to the healing process (Al-Waili et al. 2015; Martinello and Mutinelli 2021). Additionally, BV has demonstrated antibacterial, antifungal, and antiviral properties in several studies, especially Gram-positive bacterial species exhibited higher sensitivity toward bee venom (Attalla et al. 2007). In this regard, melittin and secapin are responsible peptides affecting the microorganism's infection. Accordingly, the main mechanism of the BV antibacterial effect relates to its capability in channel formation among the organism cell membrane (Kurek-Górecka et al. 2020). In an experiment on six adult rabbits infected by MRSA (methicillin-resistant *Staphylococcus aureus*) and received honey bee venom directly by a bee sting, the effect of bee venom inhibition on MRSA as one of the most resistant bacterial species was observed (Baqer and Yaseen 2018a). Also, the antibacterial and anti-inflammatory effect of topical application of bee venom and melittin on skin lesions caused by *Streptococcus pyogenes* as a prevalent microorganism found in several skin infections has been proven by Seongjae Bae et al. (2022). Furthermore, BV affects the various stages of wound healing: increasing keratinocyte migration, improving re-epithelialization, wound contraction, higher granulation, accelerating wound closure, and tissue regeneration. Some experiments have also proven its analgesic properties (Han et al. 2011; Al-Waili et al. 2015).

In a study conducted by Hozzein et al., it was found that the use of honey bee venom in diabetic wound model mice led to the promotion of angiopoietin_1/Tie2 signaling and Nrf2 expression, both of which play important roles in neovascularization. Each diabetic mouse was wounded to create 8 mm wounds and treated with 200 (μg/kg)/wounded area/day for 15 days (Hozzein et al. 2018). However, in another study in 2010, 5×5 mm full-thickness wounds in mice were treated with 1 μg. gauze^{-1} for 7 days. The results indicated a decrease in vascular endothelial growth factor (VEGF) level, vascular number, and size following bee venom treatment (Han et al. 2011; Hozzein et al. 2018).

Given the significance of expediting wound-healing processes and minimizing patient complications, various therapeutic approaches have been utilized to prevent microbial contamination. BV possesses unique properties and is non-toxic in standard therapeutic doses, allowing it to be used in various formulations, including wound dressings. Thus, this natural material could effectively treat various wounds, particularly diabetic and burn wounds prone to extensive microbial contamination. Figure 2 provides an overview of the cosmetic applications of bee venom, including its potential use in wound healing (El-Wahed et al. 2021).

Propolis Propolis is a complex mixture of beeswax, digestive enzymes, and various botanical exudates, including

Fig. 2 Wound healing as a cosmetic application of bee venom (reprinted with permission from (El-Wahed et al. 2021))

lipophilic plant secretions and herbal extracts. Honey bees produce this resinous substance to protect their hives from external threats and microbial contamination (Al-Waili et al. 2015; Martinotti and Ranzato 2015). Propolis possesses several biological and pharmacological properties, making it a promising wound healing agent. One of the active components of Propolis, caffeic acid, exhibits anti-inflammatory effects and inhibits matrix metallopeptidase 9 (MMP-9), an enzyme responsible for collagen breakdown. Propolis also can increase collagen deposition in wound sites via TGF-β1/Smad2,3 signaling pathway. As a result, it contributes to regulating collagen levels in repairing tissues (Martinotti and Ranzato 2015; Ernawati and Puspa 2018; Hozzein et al. 2015).

In the meantime, Propolis exhibits a notable bactericidal and bacteriostatic effect on certain bacterial species due to various biologically active ingredients such as phenolic acids. Besides, its antifungal properties have been demonstrated in numerous studies. For instance, in a 2019 study, a Propolis-coated polyurethane wound dressing was designed, and its antibacterial properties were examined. The results indicated efficacy against *Escherichia coli* (*E. coli*) and *Staphylococcus aureus* (*S. aureus*) (Ernawati and Puspa 2018; Khodabakhshi et al. 2019). In another experiment, the antimicrobial effect of a mixture of two samples of propolis extract was examined.

Consequently, *E. coli*, *S aureus*, and *Candida albicans* were found to be sensitive to the agent. Additionally, the wound-healing efficacy of the mixture was evaluated by twice daily administration of 1 ml of the mixed extract on 3 cm wounded skin of rabbits for 25 days. The results indicated an accelerated wound closure in treated groups with the mixed extracts compared to groups that received extracts individually or remained untreated (Al-Waili 2018).

Propolis is also known to improve blood supply by enhancing the expression of VEGF, an important angiogenesis induction factor (Sungkar et al. 2021). It exhibits other beneficial effects, such as stimulation of proliferation, migration, differentiation of cells, regulation of cell signaling, and analgesic effects. It also demonstrates immunomodulatory properties by controlling inflammatory factors, such as IL-1, IL-6, and TNF. Moreover, a topical formulation of Propolis has been shown to decrease the infiltration of neutrophils and inhibit elastase activity (Al-Waili et al. 2015; Martinotti and Ranzato 2015; Ernawati and Puspa 2018).

In an in vivo study conducted by Abu-Seida (2015) on five adult dogs with full-thickness 3 cm-diameter wounds, twice daily topical Propolis treatment was found to optimize re-epithelialization and wound contraction in the fifth week of treatment. The accelerated wound healing and improved contraction observed in the final therapeutic stage demonstrated the healing potential of Propolis without any noticeable side effects (Abu-Seida 2015).

In another study, Propolis was used as a constituent of a wound dressing along with cornstarch and hyaluronic acid. This dressing exhibited high antibacterial activity against *S. aureus*, *E. coli*, and *Staphylococcus epidermidis*. The formulation of this dressing could be a promising approach to accelerate the wound-healing process (Eskandarinia et al. 2019).

Researchers have developed a bilayer wound dressing composed of a polycaprolactone/gelatin (PCL/Gel) scaffold and a dense polyurethane and ethanolic extract membrane of Propolis (PU/EEP) due to the limitations of traditional one-layer dressings. PU/EEP is a top layer, protecting the wound from external contaminants and dehydration, while PCL/Gel promotes cell adhesion and proliferation. In rat models, this bilayer dressing accelerated skin wound healing (Eskandarinia et al. 2020). Thus, Propolis is a promising biomaterial for enhancing wound treatment.

Honey Honey is a natural bee product containing flavonoids, caffeic acid, sugars, and enzymes. It has received wide attention since ancient times as an effective ingredient in wound healing due to its unique properties. Honey helps control infections with a variety of mechanisms. Defensin is an antibacterial peptide found in honey, which is mainly responsible for its antibacterial effect, especially on Gram-positive strains. The acidic pH of honey, flavonoids, and hydrogen peroxide contribute to its antimicrobial activity (Almasaudi 2021). The properties of honey can vary due to differences in its ingredients. Various studies have confirmed honey's ability to enhance cell proliferation, epithelialization, granulation, contraction, and collagen production via NO induction and angiogenesis in the wound site. Honey also stimulates the secretion of pro-inflammatory cytokines, such as IL-1β, IL-6, and TNF, which helps induce inflammation as an essential step in wound healing (Iftikhar et al. 2010; Majtan 2014). Figure 3a illustrates the immunomodulatory effect of honey on cutaneous and immune cells in the wound-healing process. Depending on the wound condition, it can stimulate or inhibit the release of certain factors [cytokines, MMP-9, and reactive oxygen species (ROS)] from immune and cutaneous cells. During the inflammatory phase, honey induces the secretion of pro-inflammatory cytokines and MMP-9 during the proliferative phase. When the inflammation is uncontrolled, honey reduces inflammation time, the level of ROS, MMP-9. IL, and pro-inflammatory cytokines (Majtan 2014).

Honey can facilitate the regeneration of damaged tissue and the wound-healing process attributed to its characteristic features, such as pH modulation, anti-inflammatory activity, and regulation of ROS generation. These aspects can contribute to the four stages of wound healing (i.e., hemostasis, inflammation, proliferation, and tissue remodeling), as depicted in Fig. 3b (Mieles et al. 2022). Owing

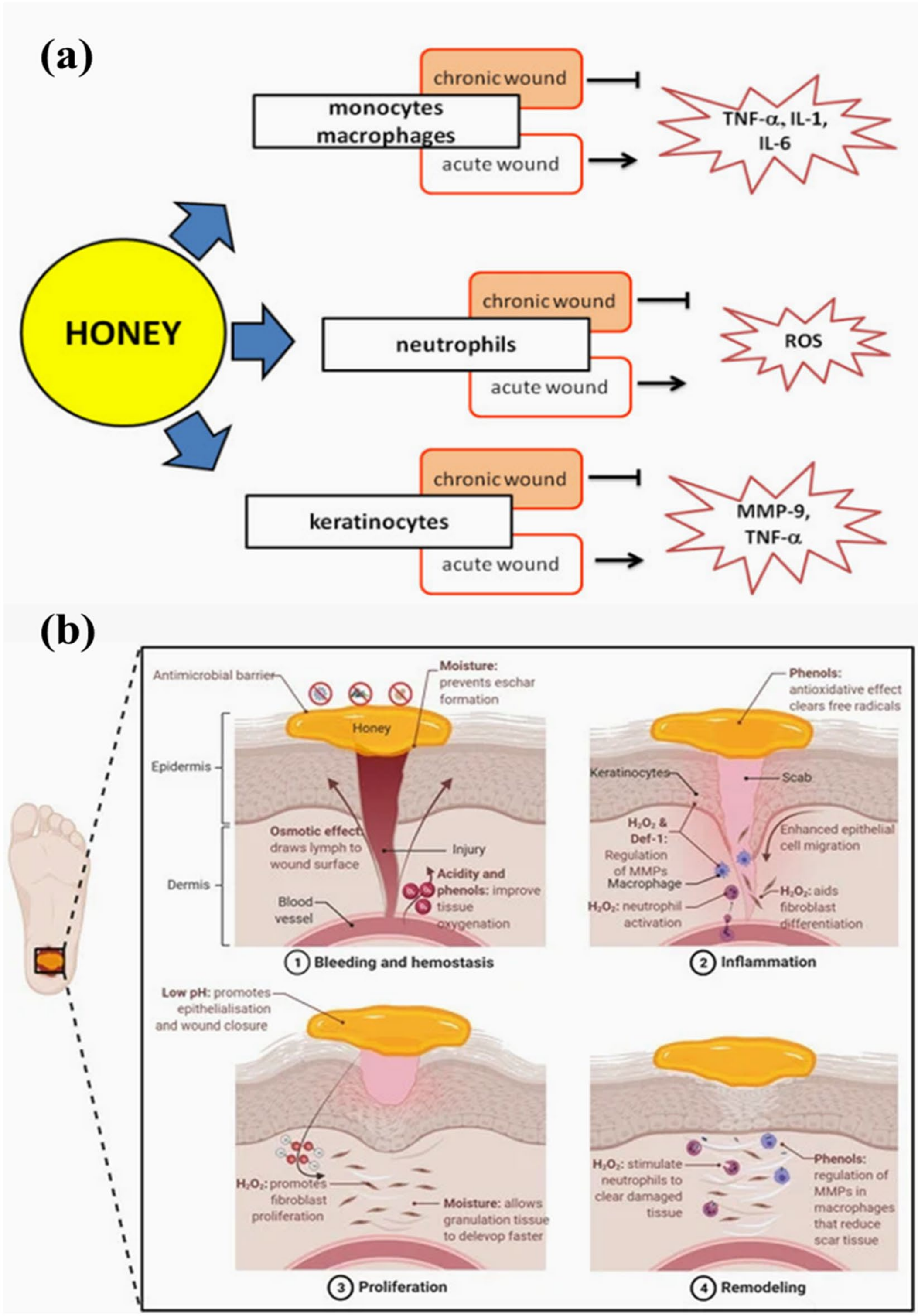


Fig. 3 (**a**) The immunomodulatory role of honey in wound healing [reprinted with permission from (Majtan 2014)]. (**b**) The mechanism representing how honey affects each stage of the wound-healing process [reprinted with permission from (Yupanqui Mieles et al. 2022)]

to its diverse mechanisms of action, honey has been extensively employed as a natural ingredient in various wounds. It is crucial to mention that, due to the possibility of some natural sources of honey contamination with harmful microorganisms such as *Clostridium botulinum* (Grenda et al. 2018), quality control and standardization of the used samples may be necessary during the experiment and product development.

Chitosan

Chitosan is a naturally occurring polymer derived from chitin, a structural component of fungal cell walls, crustacean exoskeletons, mollusk shells, and some aquatic organisms (Bano et al. 2017; Dai et al. 2011). It is a pseudo-natural polymer composed of β-(1→4)-link-D-glucosamine and *N*-acetyl-D-glucosamine. The preparation of chitosan from chitin can be achieved through alkaline deacetylation or enzymatic deacetylation, as depicted in Dai et al. (2011).

Due to its biocompatibility, biodegradability, nontoxicity, non-allergenicity, and ability to form hydrogels, chitosan is a suitable candidate for a variety of wound dressings and pharmaceutical preparations (Bano et al. 2017; Dai et al. 2011; Liu et al. 2018). Chitosan's antimicrobial properties make it a promising material for biomedical applications, particularly in wound healing. The extent of chitosan's antimicrobial activity depends on its molecular weight and degree of deacetylation (DDA) (Ke et al. 2021). Chitosan can accelerate wound healing by affecting the hemostasis phase via clotting blood and its antimicrobial, anti-inflammatory, and antioxidant properties. In addition, chitosan has low immunogenicity and can act as a tissue adhesive (Moeini et al. 2020).

Wound infection represents a significant obstacle in the wound-healing process. Researchers have investigated the potential of sulfonated chitosan (SCS) as an antimicrobial agent to overcome this issue. In a Zhimin Sun et al.'s study, SCS was synthesized and tested against various microorganisms. The results demonstrated the effectiveness of SCS against *E. coli* and *S. aureus* and its antifungal activity against *Arthrinium sacchari* and *Botrytis cinerea* (Sun et al. 2017). The antimicrobial activity of chitosan is attributed to multiple mechanisms, including the interaction of chitosan electric charges with bacterial membranes, inhibition of mRNA synthesis and expression of bacterial proteins, and increasing membrane permeability through a pH and dose-dependent mechanism (Dai et al. 2011; Sun et al. 2017). Therefore, SCS can be an effective antimicrobial agent in wound-healing applications.

The regulation of inflammation is a critical aspect of the wound-healing process, particularly during the granulation stage. Chitosan has been shown to modulate the inflammatory pathway, regulating certain cytokines, inflammatory mediators, and immune cells. Chitosan helps regulate the pathway of inflammation and consequently granulation thanks to modulation of the secretion of certain factors, such as IL-1β, IL-8, transforming growth factor β (TGFβ), prostaglandin E2 (PGE2), activating macrophages, increasing infiltration of neutrophils, and raising platelet-derived growth factor (PDGF) levels. Additionally, chitosan affects fibroblast proliferation, angiogenesis, and re-epithelialization, improving tensile strength by increasing collagen types I and III levels. Furthermore, chitosan has been reported to possess analgesic properties, which can alleviate pain associated with wounds by blocking the end receptors. It also provides a moist and oxygen-permeable environment supporting wound repair (Bano et al. 2017; Dai et al. 2011; Liu et al. 2018).

Moreover, chitosan-based hydrogels have gained significant attention as wound dressings due to their excellent biocompatibility and antioxidant properties. For instance, a hydrogel has been developed as effective bio-adhesive wound dressings to promote healing and hemostasis in Sprague Dawley (SD) rats (Sun et al. 2022). Given its versatile properties, chitosan has been utilized as a promising component in combination with natural and synthetic materials for repair. Some of the applicable forms of chitosan with wound regenerating capability are demonstrated in Fig. 4.

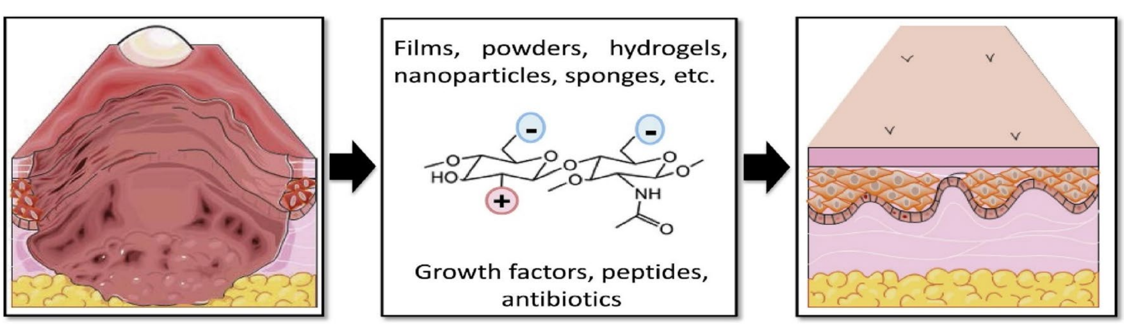


Fig. 4 The applicable forms of chitosan in combination with natural and synthesis material as the wound repairing agents [reprinted with permission from (Patrulea et al. 2015)]

Silk

Silk is a protein substance produced by specialized epithelial cells in certain organisms, such as silkworm *Bombyx mori* and spider. Silk composition comprises two main protein components: fibroin and sericin. Fibroin is a fibrous structural protein, whereas sericin is a water-soluble protein with film-forming properties (Ersel et al. 2016; Suarato et al. 2018). Silk fibroin can exist in various forms, such as silk fibroin solution, nanofibrous scaffolds, hydrogels, and sponges/blend films (Vidya and Rajagopal 2021).

In a recent study by Chouhan et al., fibroin hydrogels were injected into third-degree burn wounds to evaluate their efficacy in wound healing. The results showed that type I and III collagen levels increased significantly on days 7–21 and day 14 post-treatment, respectively. The re-epithelialization and regeneration processes were also accelerated in the treated group compared to the control group. The regulation of immune cell function by fibroin hydrogels led to the control of inflammatory responses. By the 21st day, the wound was completely closed in the treated group, while the control group still had a 30% open wound (Chouhan et al. 2018).

Hydroxyproline is a precursor of collagen, and its production is an important indicator of collagen synthesis. Sericin, a water-soluble protein component of silk, has been shown to increase the production of hydroxyproline and consequently enhance collagen levels. Additionally, sericin plays a role in collagen synthesis by regulating nitric oxide (Ersel et al. 2016; Aramwit et al. 2013). These findings suggest that silk-based hydrogels have the potential to improve wound-healing outcomes by enhancing collagen synthesis and controlling inflammatory responses.

Silk has been found to play an important role in wound healing through various mechanisms, including the stimulation of fibroblast and keratinocyte proliferation and migration, enhancement of granulation, and reduction of scarring by promoting the formation of thick tissue. Additionally, sericin has been demonstrated to affect angiogenesis and improve vascular survival in wounds, particularly burn wounds. It has also been shown to possess potent antioxidant properties by reducing peroxyl and malondialdehyde (MDA) levels and activating antioxidant enzymes, such as glutathione peroxidase, superoxide dismutase (SOD), and catalase (Ersel et al. 2016; Chouhan et al. 2018; Aramwit et al. 2013).

Silk films have been utilized as an alternative treatment for skin wounds, demonstrating a significantly faster wound-healing rate and lower inflammatory response than conventional wound dressings (Sugihara et al. 2000). Padol et al. have reported that silk fibroin film, when integrated with epidermal growth factor, shows exceptional wound-healing performance (Liu et al. 2019). Notably, the efficacy of both

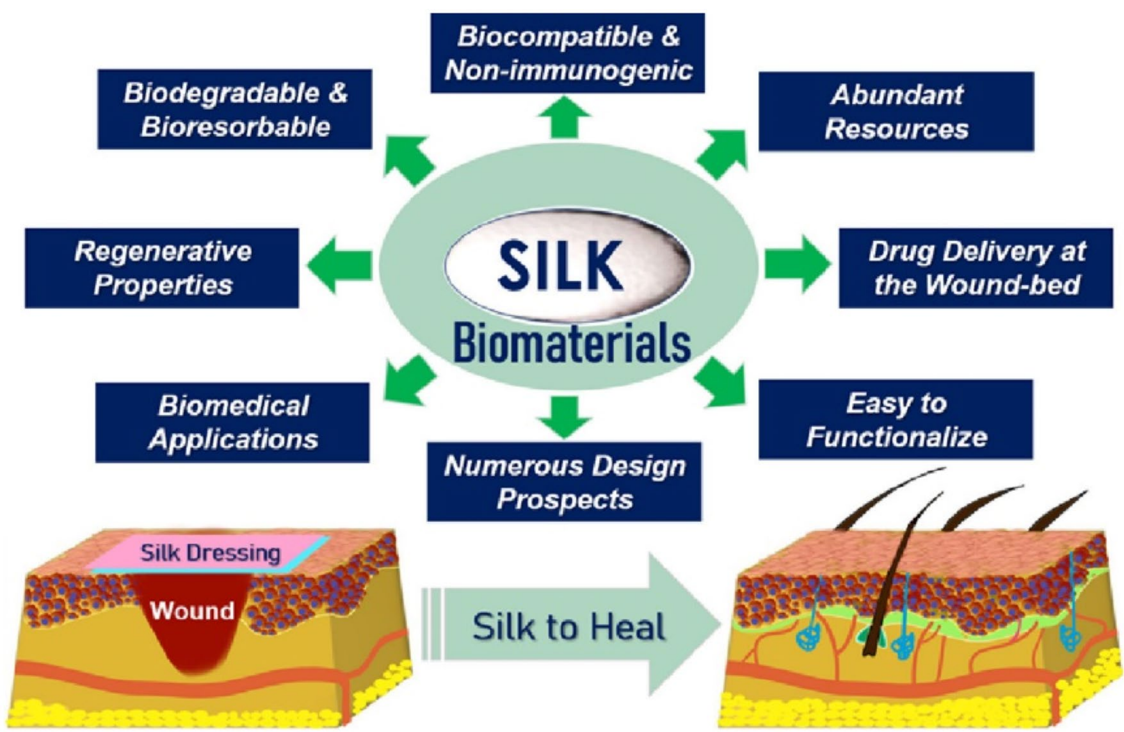


Fig. 5 Characteristics of silk biomaterials applied in wound healing [reprinted with permission from Chouhan and Mandal 2020; Naskar et al. 2014)]

silk components in wound healing has been well documented. Due to its restorative properties, moisture retention capability, and favorable environment, fibroin is widely used as a base material for various medical products, scaffolds containing diverse drugs, as well as natural and biochemical materials (Suarato et al. 2018). The characteristics of silk biomaterials are illustrated in Fig. 5 (Chouhan and Mandal 2020).

Medical plants for wound-healing

Calendula officinalis

Calendula officinalis (*C. officinalis*) belongs to the Asteraceae family that has been traditionally used to treat various skin conditions, including surgical wounds, dermatitis, burns, and skin irritation caused by chemotherapy (Arora et al. 2013). The topical application of *C. officinalis* extract is effective in different stages of wound healing, including the stimulation of collagen synthesis and tissue regeneration, as evidenced by increased levels of hydroxyproline and hexosamine. In addition, it has been revealed that the extract affects the inflammatory phase of wound healing by stimulating the NF-κB and by raising IL-8 levels, at the transcriptional and protein phase, in keratinocytes (Nicolaus et al. 2017; Preethi and Kuttan 2009). These findings highlight the potential therapeutic benefits of *C. officinalis* extract in wound healing.

Moreover, C. officinalis enhances wound healing by promoting tissue contraction, proliferation, and migration of dermal fibroblasts. Additionally, the extract may stimulate angiogenesis, promoting the formation of new blood vessels for tissue nourishment (Parente et al. 2012). Notably, a study conducted on patients with acute hand wounds demonstrated that topical application of *Calendula officinalis* extract accelerated wound healing by promoting faster epithelialization (Giostri et al. 2022).

In addition, *Calendula officinalis* has been incorporated into a chitosan-based scaffold, which has shown improved antibacterial activity against Gram-positive and Gram-negative bacteria. The scaffold enhanced cell proliferation and wound-healing capabilities (Kharat et al. 2021).

Furthermore, beyond its topical application, calendula extract has also been investigated for its potential systemic effects on wound healing. In an in vivo study using a burn wound model, 200 mg/kg oral administration of calendula extract for 10 days in rats increased hydroxyproline and hexosamine, suggesting improved collagen synthesis in the wound. Additionally, the extract showed inhibitory effects on the rise of acute-phase proteins, indicating reduced undesirable systemic damage caused by burn injuries. The observed decrease in liver enzyme levels further supports this effect. Calendula extract also exhibits antioxidant properties, as evidenced by an increase in glutathione, superoxide dismutase (SOD), and catalase levels and a decrease in lipid peroxidase levels (Chandran and Kuttan 2008).

In an experiment, the function of two types of Calendula officinalis extracts on the wound-healing process was evaluated. *C. officinalis* extracts can be obtained using different solvents, such as n-hexanic (HE), ethanolic (EE), and aqueous (AE) extracts, followed by solvent removal. HE and EE extracts have been found to modulate the inflammatory phase of wound healing by activating the NF-κB transcription factor and subsequently inducing the transcription of the pro-inflammatory chemokine IL-8 (Nicolaus et al. 2017).

C. officinalis extract has been shown to benefit wound healing when administered topically or orally. This extract has been used to treat various types of wounds, including diabetic foot ulcers, venous leg ulcers, and burns.

Matricaria recutita

Matricaria recutita, commonly known as chamomile, is another member of the Asteraceae family widely used in pharmaceutical, cosmetic, and traditional medicine. Chamomile extract and essential oil contain various active ingredients which significantly affect different stages of wound healing and have been extensively studied. The essential oil of chamomile, which consists of approximately 52 compounds, exhibits antibacterial properties. Its inhibitory effect on *S. aureus* and *Listeria monocytogenes* has been confirmed, while no significant effect has been observed on *Pseudomonas aeruginosa* (Stanojevic et al. 2016). The antifungal activity of chamomile extract has also been confirmed in some species, especially *Candida albicans* (Wagih Abd ElFattah and El 2014).

In addition to its antibacterial properties, chamomile extract can effectively promote wound closure. Corticosteroids are the most common drugs to control inflammation and accelerate wound healing. In a study by Martins et al. (2009), the effects of corticosteroids and chamomile extract were compared by twice daily administration of chamomile extract, triamcinolone cream, clobetasol paste, and cream in 3 cm tongue wound in rats. Consequently, chamomile extract was found to be more effective and had fewer side effects. However, cell viability was lower in chamomile than in corticosteroids, although this effect was negligible (Martins et al. 2009). Burn wounds can cause long-term local and systemic injuries, including inflammation, widespread infections, and reperfusion injury to the patient. Chamomile extract has been found to effectively accelerate the healing of such wounds, indicating its potential as a therapeutic agent (Jarrahi 2008).

Chamomile, besides its remarkable effects on the skin, has also been found to have regenerative properties on mucosal tissues, such as oral ulcers. An investigation conducted on rats demonstrated that topical application of an ointment containing 10% chamomile extract resulted in an increase in hydroxyproline content and accumulation of fibroblasts at the wound site, leading to a rise in collagen levels and improvement in epithelialization, thus effectively accelerating the healing process. Furthermore, the formulation alleviated painful symptoms of oral ulcers by reducing inflammation. As a result of these beneficial properties, oral products, such as mouthwashes and drops containing chamomile extract and essential oil, are extensively used (Duarte et al. 2011).

Oral administration of chamomile extract has been shown to enhance wound healing via mechanisms similar to those of topical application. In an in vivo study, an aqueous chamomile extract was orally administered to animal models with excision, incision, and dead space wounds at 120 mg/kg/day. The results indicated that the extract led to increased levels of hydroxyproline, accumulation of fibroblasts, and subsequent production of collagen, resulting in improved wound strength. Moreover, the extract promoted granulation and epithelialization, leading to accelerated wound closure. The extract's flavonoid compounds and antioxidant properties create a favorable environment for wound healing by regulating the inflammatory process and reducing immune cells at the wound site. However, the extract was ineffective against *Pseudomonas aeruginosa*, *E. coli*, and *S. aureu*s, but it had an inhibitory effect on *Cedecea neteri* (Shivananda Nayak et al. 2007).

The therapeutic potential of *Matricaria recutita* in managing diabetic wounds has also been established. Specifically, *Matricaria recutita* extract has been found to enhance wound healing by promoting fibroblast proliferation and revascularization in diabetic skin injuries

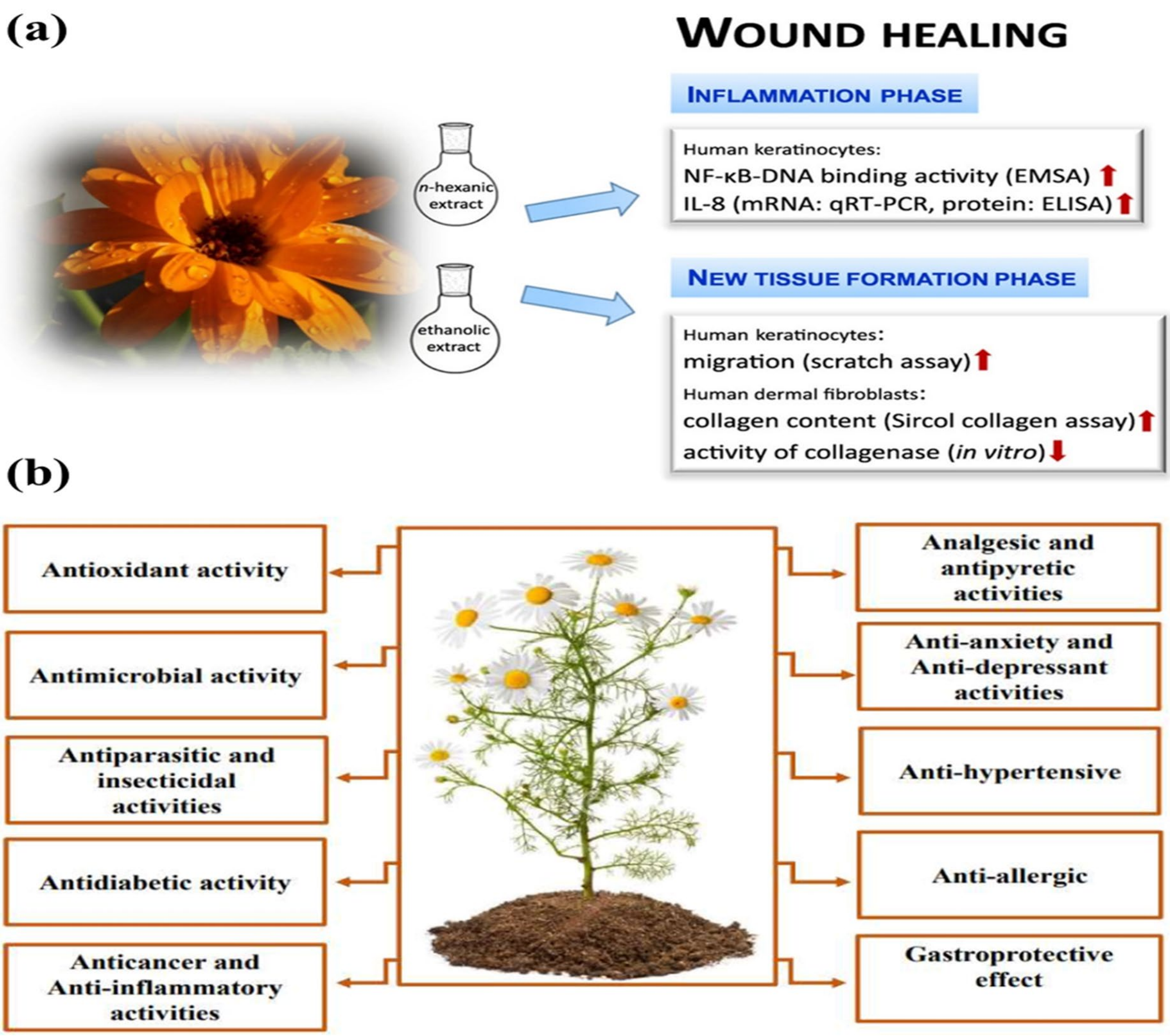


Fig. 6 (**a**) Calendula officinalis extract and their functions in the wound-healing process (reprinted with permission from (Nicolaus et al. 2017)), (**b**) Biological properties of *Matricaria recutita* process [reprinted with permission from (El Mihyaoui et al. 2022)]

(Nematollahi et al. 2019). This has been attributed to the various bioactive constituents present in the extract, such as flavonoids and terpenoids. The underlying biological properties of these constituents are illustrated in Fig. 6b (El Mihyaoui et al. 2022).

Punica granatum

Punica granatum, known as pomegranate, is a well-known and extensively used plant that exhibits unique effects through its various components, including fruit, peel, and leaves, when administered systemically and topically.

HPLC analysis of a methanolic extract of pomegranate peel has revealed the presence of effective compounds, such as punicalagin, anthocyanins, and gallic acid. In a study by Nema et al., rats with excision ulcers were treated with *Punica granatum* 50% methanolic extract ointment (10% and 15% w/w). In another study conducted by Hayouni and co-workers, the wound-healing process in animal models of guinea pigs with an excision wound was treated with 5% w/w methanolic extract-based ointment from *P. granatum L.* peels, and the results were compared to the standard placebo and untreated groups.

Both studies showed that the pomegranate extract significantly improved wound contraction, granulation, tissue maturation, tensile strength, scar reduction, and integrity. Furthermore, the extract stimulated protein expression in the treated group's granulation tissue, indicating a proliferative effect. Histological and structural examination revealed increased thickness of the keratin layers, granulation, epidermis, and basal layer, possibly due to the extract's effect on collagen production, interlayer connections, and improved epithelialization. Compared with the control group, the extract also induced angiogenic effects in the treated group, leading to better tissue nourishment. The extract demonstrated antimicrobial properties against *Trichophyton rubrum* as one of the most resistant bacteria, while *Candida albicans*, *Pseudomonas aeruginosa*, and *E. coli* were less sensitive (Hayouni et al. 2011; Nema et al. 2013).

In 2017, Nirwana et al. conducted an in vivo experiment to evaluate the effect of pomegranate extract in a gel formulation on post-tooth extraction wounds. The treated group exhibited increased fibroblast proliferation and improved collagen production at the wound site. The researchers also investigated the effect of pomegranate fruit extract on growth factors, which indicated that the extract stimulates the production of vascular endothelial growth factor (VEGF) from macrophages and fibroblasts, thereby promoting angiogenesis as a vital step in the wound-healing process. Moreover, the extract enhances the angiogenic effect of VEGF by elevating platelet-derived growth factor (PDGF) levels and actively recruits immune cells and fibroblasts to the wound site (Nirwana et al. 2017).

Recently, the application of plant extracts in modern technologies has become increasingly popular. For instance, Amal and colleagues developed a novel drug delivery system by loading pomegranate pericarp extract into a chitosan-collagen-starch membrane (Amal et al. 2015). Moreover, a combination of two plant extracts, *Matricaria chamomilla* and *Punica granatum L*, has been proposed for wound healing. The combination is designed to leverage the astringent activity of *Punica granatum* and the anti-inflammatory and antioxidant activities of *Matricaria chamomilla*, which have been traditionally used as natural remedies for skin problems, including wound healing (Niknam et al. 2021).

Rosmarinus officinalis

Rosemary (*Rosmarinus officinalis L.*) is a perennial shrub belonging to the Lamiaceae family, originating from Mediterranean countries, although other species can be found in other regions (Takaki et al. 2008; Yilmaz et al. 2012). The essential oil and extract of rosemary have been extensively studied in both total and fragmented forms for their potential therapeutic effects on various local and systemic health issues. The reported effects of rosemary include noteworthy outcomes.

In 2007, Takaki et al. evaluated the anti-inflammatory and analgesic effects of rosemary extract through carrageenan-induced paw edema, acetic acid-induced writhing, and hot plate tests. The study assessed the safety of rosemary extract at doses of 1000 mg/kg as intraperitoneal (IP) and 3000 mg as oral administration (PO) and found no adverse effects on the cell. Oral administration of essential oils containing myrcene and linalool helped reduce inflammation and had antinociceptive properties, with a better response observed in animal models exposed to chemical and environmental pain compared to thermal and central stimuli. The mechanism of action was attributed to the prevention of histamine release, which helped reduce the irritant effects of injuries (Takaki et al. 2008).

In addition to rosemary essential oil, this plant extract also benefits wound healing. The efficacy of rosemary extract and essential oil in wound healing is evidenced by their ability to enhance better epithelialization, accelerate wound size reduction, improve neovascularization, and provide an infection-free environment due to their antibacterial properties. Moreover, rosemary extract stimulates the proliferation of fibroblasts and increases collagen synthesis, as well as forming an extracellular matrix, better contraction, and improved granular tissue regeneration. Notably, the extract promotes the formation of mature keratin, epidermis, and dermis layers with proper thickness, highlighting the unique healing effects of rosemary extract in normal and diabetic wounds. In diabetic wounds, where the lack of neutrophils and macrophages at the wound site delays the inflammatory process and coagulation, the immuno-regulatory effect of rosemary extract and essential oil contributes to the proper presence of immune cells at the wound site (Abu-Al-Basal 2010; Yilmaz et al. 2012).

In recent years, there has been a growing interest in modern drug delivery systems, leading to the design of encapsulated rosemary essential oil in nano-dimensional lipid carriers. The gel containing this compound has been shown to possess anti-inflammatory and antibacterial activities against several pathogens, including *Staphylococcus epidermidis*, *S. aureus*, and *Staphylococcus monocytogenes*. The granulation process of the tissue was significantly accelerated with the application of this compound, as evidenced by the completion of the process on day 12 compared to day 16 for mupirocin. The levels of IL-3, IL-10, and VEGF were higher in the treated groups, indicating the stimulatory effect of the extract on fibroblast migration and collagen production. These findings suggest that the encapsulated rosemary essential oil in nano-dimensional lipid carriers can be a promising drug delivery system for wound-healing applications (Khezri et al. 2019).

Recent studies have investigated the effectiveness of *Rosmarinus officinalis* in wound healing. One experiment evaluated the effect of *Rosmarinus officinalis* essential oils on cutaneous burn injury in rats and found a significant healing activity during the proliferative phase of wound healing (Belkhodja et al. 2020). In another study, *Rosmarinus officinalis* hexane extract was evaluated for its photoprotective, anti-inflammatory, and antioxidant properties. In vitro investigations demonstrated the extract's antioxidant and anti-aging properties and its potential in wound healing. Furthermore, the extract showed a good safety profile, high efficiency, and cost-effectiveness, making it a promising candidate for further development (Ibrahim et al. 2022). Given the effectiveness of rosemary in different stages of wound healing and its potential to reduce wound complications, combining its extract and essential oil may be a beneficial addition to wound-healing products.

Green tea

Camellia sinensis, commonly known as green tea, belongs to the Theaceae family and contains a variety of components, including polyphenols, polysaccharides, flavonoids, and catechins (as the main antioxidant found in green tea extract), along with over 100 other substances (de Almeida Neves et al. 2010; Shahrahmani et al. 2018). Topical application of green tea extract on animal models with surgical wounds has shown significant effects on wound-healing processes, resulting in shorter wound repair times and improved healing quality, likely due to the improved epithelialization and granulation process and the extract's antioxidant properties. The extract also enhances angiogenesis, increasing blood supply and tissue nutrition. It should be noted that the systemic use of the extract through oral administration has shown similar effects on the healing process (de Almeida Neves et al. 2010; Asadi et al. 2013).

In addition to conventional applications of plant extracts, Park et al. conducted a study in 2014 in which green tea extract was loaded onto microneedles containing hyaluronic acid to assess its antibacterial and restorative properties. An in vivo examination was performed using Sprague–Dawley rat models with full-thickness excisions, followed by induction of microbial infection with *E. coli*, *Salmonella typhimurium*, *Pseudomonas putida*, *Bacillus subtilis*, and *S. aureus*, and treatment was performed with microneedles containing hyaluronic acid and green tea extract. The concentration-dependent inhibitory effect of the extract was observed on species, such as *Bacillus subtilis*, *P. putida, S. aureus*, and *S. typhimurium*, indicating the antibacterial effect of green tea. Catechins are the main active ingredients responsible for this property, which was also confirmed by the in vitro study. The antibacterial effects are exerted by cell wall and membrane destruction (Park et al. 2014).

Epigallocatechin gallate (EGCG), a polyphenol found in high abundance in green tea leaves, has been shown to possess notable anti-inflammatory, antioxidant, and antimicrobial properties. Figure 7a illustrates the diverse forms in which green tea plants have been utilized for wound-healing applications (Xu et al. 2021). Green tea possesses several unique properties, including but not limited to its antioxidant, anti-inflammatory, and healing properties, making it a favorable option for the treatment of various types of wounds.

Hypericum perforatum

Hypericum perforatum, commonly known as St. John's Wort, is a member of the Hypericaceae family with notable properties have made it a popular ingredient in modern-day products and traditional medicine. The effectiveness of this plant extract in treating a range of skin conditions, including chronic wounds, eczema, post-surgical wounds, gastrointestinal wounds, and hemorrhoids, has been demonstrated in numerous studies (Altıparmak and Eskitaşçıoğlu 2018; Yadollah-Damavandi et al. 2015).

Several studies have investigated the efficacy of *H. perforatum* extracts in treating wounds of different types, including chronic wounds and diabetic wounds, using various extract formulations, such as hydroalcoholic, oily, and direct application, and multiple carriers. The results of these studies demonstrate the restorative properties of *H. perforatum* extracts and their ability to accelerate the healing process by promoting fibroblast migration and accumulation in the dermis. Additionally, treatment with *H. perforatum* extracts has been found to enhance wound contraction and tensile

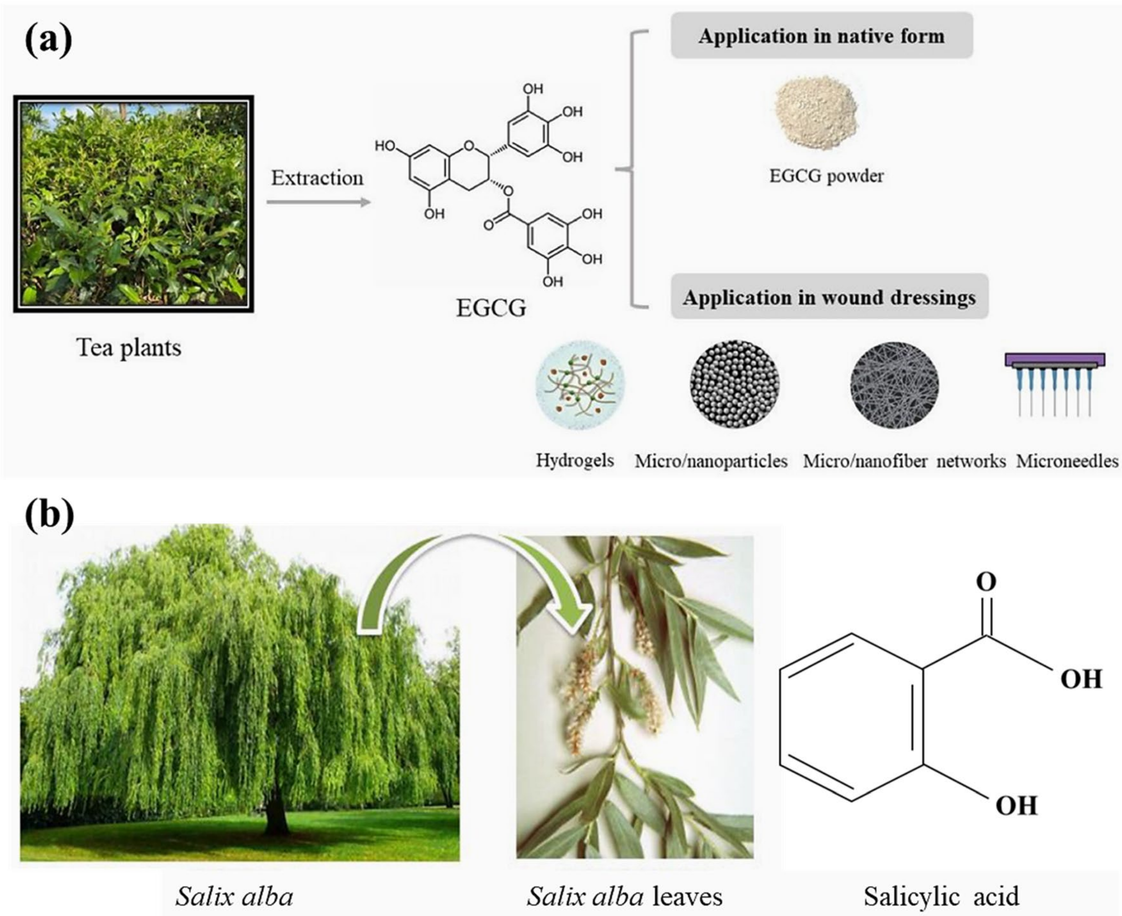


Fig. 7 (**a**) Application of green tea plant in various formulations for wound-healing applications [reprinted with permission from (Xu et al. 2021)]. (**b**) Salix alba, leaves, and main compound (salicylic acid) [reprinted with permission from (Qureshi et al. 2015)]

strength, which may be attributed to increased hydroxyproline levels and collagen fiber accumulation in the wound area (Yadollah-Damavandi et al. 2015; Altıparmak and Eskitaşçıoğlu 2018; Süntar et al. 2010).

In the meantime, angiogenesis, a crucial process in wound healing, can be effectively promoted by *H. perforatum* extract. In a 2017 investigation by Altıparmak and Eskitaşçıoğlu, the regulatory effect of *H. perforatum* extract on angiogenesis was studied in a diabetic rat model. During the first week of treatment, the extract was found to accelerate the process of angiogenesis, thereby promoting wound healing. However, excessive angiogenesis can lead to abnormal granulation and improper repair. To prevent this phenomenon, the extract was found to reduce the rate of angiogenesis in the third week. In addition, oral administration of the extract was found to have a more significant effect on all of the stages mentioned above compared to topical application. Notably, the efficacy of the extract was found to be significantly higher than that of olive oil (Yadollah-Damavandi et al. 2015; Altıparmak and Eskitaşçıoğlu 2018).

H. perforatum has been found to have anti-inflammatory, antibacterial, and wound healing properties. The plant extract has been demonstrated to regulate the inflammatory process, leading to shorter periods of inflammation comparable to indomethacin, a reference compound. Moreover, the extract exhibited a regulatory effect on diabetic wounds, which are characterized by a deficiency in the inflammatory process and impaired healing (Altıparmak and Eskitaşçıoğlu 2018; Süntar et al. 2010). *H. perforatum* also exerts an antibacterial effect and is effective against infections caused by Gram-positive and Gram-negative bacteria, thanks to its content of hypericin, hyperforin, and other compounds (Cecchini et al. 2007).

In a comparison study, the extract of *H. perforatum* was safer and more effective than povidone-iodine, tincture benzoin, and tretinoin in promoting surgical wound healing. *H. perforatum* extract significantly increased re-epithelialization and collagen accumulation. These results confirm the potential of *H. perforatum* as a promising natural remedy for wound healing (Yalcınkaya et al. 2022).

Table 1 The summary of clinical studies considering the wound-healing properties of natural agents

Type of study	Number of patients	Wound model	Natural agent	Administration	Duration of experiment	Control group (number of patients)	Outcomes	References
Clinical trial study	24	Diabetic foot ulcer	Propolis aqueous liquid form	A thin layer of the solution once daily	6 weeks	Diabetic foot ulcer, recently treated (84)	- Accelerated wound closure - Reduced bacterial colony forming - Faster reduction in MMP-9 level	Henshaw et al. (2014)
Randomized controlled trial	20	Diabetic foot ulcers	Propolis (topical 5% ointment)	A thin layer of 5% cream twice daily	4 weeks	Diabetic foot ulcers receive the same treatments except for propolis. (20)	- Decreased neutrophil counts and ESR level - Accelerated wound closure - No significant effect on ulcer discharge and erythema	Afkhamiz Adeh et al. (2018)
Randomized placebo-controlled clinical study	31	Diabetic foot ulcers	Propolis (Spray 3%)	A layer to cover the wound once daily	8 weeks	Diabetic foot ulcer receiving the same treatments except for propolis (11)	- Improved wound closure - Noticeable effect on the presence of fibrosis and connective deposit - Decreased TNF-α levels and - Not a significant changes in IL-10 level but increased level in net change analysis	(Mujica et al. 2019)
Randomized controlled clinical Phase II	30	Chemotherapy-related oral mucositis	1-Honey 2-Mixture of honey, olive oil–Propolis extract, and beeswax	Group1: 0.5 g honey/kg of topical honey Group2: 0.25 g/kg of a 4:2:1 mixture of honey, olive oil– propolis extract and beeswax 3 times daily	10 days	Topical benzocaine 7.5% (30)	- Accelerated recovery time in patients treated with both compounds compared with the control group	(Abdulrhm an et al. 2012)

Table 1 (continued)

Type of study	Number of patients	Wound model	Natural agent	Administration	Duration of experiment	Control group (number of patients)	Outcomes	References
Randomized, double-blinded clinical study	32	Diabetic foot ulcers	Manuka honey (Manuka honey impregnated dressings)	Initially, treatment occurred once daily and followed based on the wound condition	16 weeks	Patients receiving saline soaked gauze dressings (31)	- Proper antiinfection action - Reduction in antibiotic therapy in the treatment group	(Kamaratos et al. 2014)
Randomized clinical study	52	Symmetrical incisions wound caused by surgery	Honey (dressing containing honey)	-	-	-	- Reduced scar formation - Reduced erythema - Lower complication - Reduced risk of wound dehiscence	(Goharshenasan et al. 2016)
Double-blinded randomized clinical trial	40	Episiotomy wound in nulliparous women	Honey (honey cream 30%))	Once daily	10 days	Patients received phenytoin cream 1% (40) and placebo (40)	- Significantly decreased redness, edema, discharge from the wound, ecchymosis, and approximation (REEDA scale) in the treatment group - More descending	(Lavaf et al. 2017)
Prospective interventional clinical study	20	Non-healing wounds	Honey (dressing containing manuka honey 99% and manuka oil 1%)	-	40 days	Patients received povidoneiodine, nanocrystalline silver, or hydrogel (20)	- Accelerated wound healing -Lower pain score	(Zeleníková and Vyhlídalová 2019)

Table 1 (continued)

Type of study	Number of patients	Wound model	Natural agent	Administration	Duration of experiment	Control group (number of patients)	Outcomes	References
Randomized clinical study	34 advanced-stage cancer patients	Malignant Wounds	Honey (Manuka honeycoated bandages)	a dress changed every 2–3 days	4 weeks	nanocrystalline silvercoated bandages (35)	- Proper wound size reduction but not statistically significant in comparison with silver-coated bandages - Superior wound cleanliness - No significant effect on malodor, exudation, and wound pain compared with patients treated with silver-coated bandages	(Lund-Niel sen et al. 2011)
Randomized comparative clinical study	37	Burn wound	Honey (cream)	A layer of cream once a day	-	Patients received silver sulfadiazine cream(41)	- Accelerated wound healing - Controlled infection	(Baghel et al. 2009)
Randomized clinical study	45	Unhealed chronic wounds, including pressure ulcers, vascular ulcers, diabetic foot ulcers, and wounds with minor infections or at risk of infection	Chitosan (Wound dressing)	A 10×10 wound dress changed every 3-4 days	4 weeks	Patients receiving Vaseline gauze (45)	-Superior wound area reduction -Lower pain during dressing removal -Lower average pain level -Lower wound depth -Less wound exudate -Suitable healing in infected wounds	(Mo et al. 2015)
Single-blind, prospective, randomized clinical study	31	Loop electrosurgical excision wound	Chitosan (tampon)	-	4 weeks	Patients receiving general tampon (31)	- Lower blood clotting index - Improved blood coagulation - Lower vaginal discharge - Lower complication after surgery	(Chong et al. 2020)

Table 1 (continued)

Type of study	Number of patients	Wound model	Natural agent	Administration	Duration of experiment	Control group (number of patients)	Outcomes	References
Randomized, double-blinded, standard controlled clinical study	29	Second degree burn wound	Silk sericin (mixture of sericin and silver zinc sulfadiazine cream)	once daily	Until the complete wound healing	Patients receiving silver zinc sulfadiazine cream	-Accelerated re-epithelialization, wound size reduction, and complete healing -Shorter hospitalization period -Slightly lower pain score but not significantly different -Infection-free wound healing	(Aramwit et al. 2013)
Randomized, controlled, matched-pair clinical study	110	Split-Thickness Skin Graft	Silk (sericinreleasing wound dressing)	A 4×4 cm wound dress	13–16 days	-	- Accelerated wound healing - Lower pain score	(Siritientong et al. 2014)
Prospective nonrandomized controlled clinical study	57	Chronic venousle gulcers	*Calendula officinalis* (spray of Plenusder max)	Twice daily	30 weeks	Patients received collagenase (0.6U/g), chloramphenicol (10mg/g), and 1% silver	- A tibacterial activity - Reduced pain, erythema, and edema - Improved contraction and reepithelialization - Accelerated wound closure	(Buzzi, De Freitas, and de Barros Winter 2016)
The prospective, descriptive pilot study	84	Diabetic foot ulcers	*Calendula officinalis* (hydroglycolic extract spray)	Twice daily	30 weeks	-	- Accelerated wound closure -Improved contraction and reepithelialization	Buzzi, de Freitas, and Winter 2016)
Controlled Clinical Study	36	Skin lesion of colostomy	Chamomile	Twice daily	28 days	Patients received Hydrocortisone 1% ointment (36)	Faster wound closure	(Charousa ei, Dabirian, and Mojab 2011; Afkhamizadeh et al. 2018)

Table 1 (continued)

Type of study	Number of patients	Wound model	Natural agent	Administration	Duration of experiment	Control group (number of patients)	Outcomes	References
triple-blind randomized clinical trial	40	Episiotomy wound	Rosemary (cream)	2 cm of cream twice daily	10 days	Patients received placebo cream	-Redness, edema, ecchymosis, discharge, and approximation of the wound edges were lower in treated group.	(Hadizadeh-Talasaz et al. 2022
Randomized clinical trial	35	Pressure ulcer	Rosemary (ointment)	A layer of 2–3 mm ointment once a day	7 days	position of patients changed, and the wounds were washed with normal saline (35)	-Faster wound closure -prevention of grade 1 pressure ulcer	(Parizi, Sadeghi, and Heidari 2021)
A randomized, double-blind clinical trial study	144	Cesarean wound	*H. perforatum* (ointment)	A layer of ointment 3 times a day	16 days	group1 received placebo(44) and group 2 received no interventio n (34)	- Superior wound healing - Lower scar formation - Lower vascularity of scar but not significantly different	(Samadi et al. 2010)
A double-blind clinical trial study	35	Episiotom y wound primiparous women	*Hypericum perforatum* (ointment)	1 cm of ointment twice daily	10 days	2 groups of 35 patients. group1 received placebo(Vaseline), and group 2 received no intervention	- Lower pain severity - Reduced redness and ecchymosis - Decreased wound discharge	(Hajhashemi et al. 2018)

Salix alba

Salix alba L. is a member of the Salix genus within the Salicaceae family, which includes a variety of species ranging from small shrubs to large trees up to 30 m in height (Fig. 7b). For thousands of years, *S. alba* has been used for its medicinal properties, particularly as a treatment for pain and inflammation ('Effect of Aloe Vera and Propolis on Wound Healing in Pressure Injuries' 2022). Hippocrates recommended chewing willow bark for patients with fever, inflammation, and pain, and it has continued to be used for this purpose throughout history ('The Effectiveness of Stingless Bee Honey (Kelulut Honey) Versus Gel in Diabetic Wound Bed Preparation' 2023). Despite its long-standing use, further research is still required to elucidate the medical properties of *S. alba*. Its leaves contain a variety of chemical compounds, including linolenic acid, salicyl alcohol, 4, 6-O-nonylidene, stearyl aldehyde 4-acetoxy-3-methoxycinnamic acid, galactose, and stearic acid ('Enhanced Secondary Intention Healing vs. Standard Secondary Intention Healing in Mohs Surgical Defects on the Head and Distal Lower Extremities.' 2022).

The bark of *S. alba* is used as an analgesic, anti-inflammatory, and antiseptic compound due to its salicin content, which can be converted to salicylic acid (Fig. 7b) or aspirin in the human body. Aspirin (acetylsalicylic acid) is chemically similar but not identical to salicylic acid. Salicylates cause disease resistance in plants. Salicylic acid, a mild antibiotic, antipyretic, and antifungal agent, is also considered an antioxidant and phytochemical. Salicylic acid was first extracted from *S. alba* by European chemists in 1828 and later modified to create aspirin. Salicylic acid is effective in treating pain and fever but is also sufficiently irritating in burning off warts. Embryonic *S. alba* buds exhibit strong antiseptic and disinfectant properties, making them useful in topical wound care. *S. alba* buds also demonstrate germicide and insecticide properties and protect the skin from bacterial and fungal infections when incorporated into wound dressing systems, mainly due to the components, such as salicylic acid and methyl salicylate. Additionally, topical application of *S. alba* has effectively reduced wrinkles and sagginess of the skin following wound repair (Le Son and Nguyen 2013).

Natural agents used in wound-healing clinical trial studies

Numerous clinical trials have been conducted between 2010 and 2022 to evaluate the effectiveness of natural agents in wound healing. These agents have demonstrated significant efficacy in accelerating wound closure, reducing wound size, and controlling infection and inflammation, wound discharge, and scar formation across various studies. A summary of the findings is provided in Table 1.

Conclusion and prospects

Numerous skin and wound-healing agents have been developed to improve patients' quality of life, but adverse side effects often accompany their effectiveness. Natural compounds are a promising alternative for developing wound-healing drugs due to their potential immunomodulatory, anti-inflammatory, antimicrobial, and antioxidant properties. Our investigation focused on several natural products with these properties, informed by current research articles, clinical studies, and publications from 2007 to 2022. Using natural substances in personal care products and wound treatment has a long-standing history in traditional medicine. However, the limited number of clinical trials and methodological differences necessitate comprehensive, well-designed, randomized-controlled trials with validated and consistent outcome measures to establish the efficacy of natural agents in wound healing.

Declarations

Conflict of interest The authors declare no conflict of interest.

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